

MANUAL BOOK

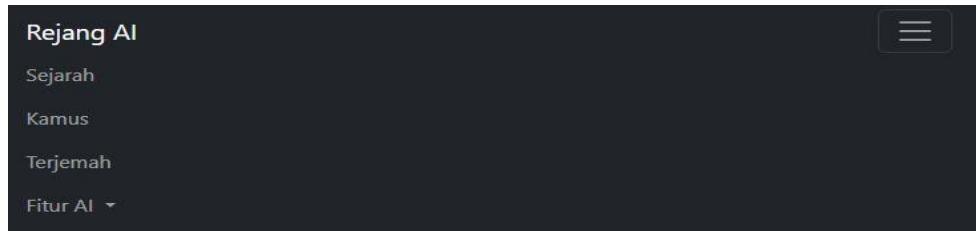
REJANG AI

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A. Interface

1) Home



Rejang AI merupakan aplikasi untuk mengenal tentang Bahasa dan Budaya Rejang yang terdapat di Provinsi Bengkulu

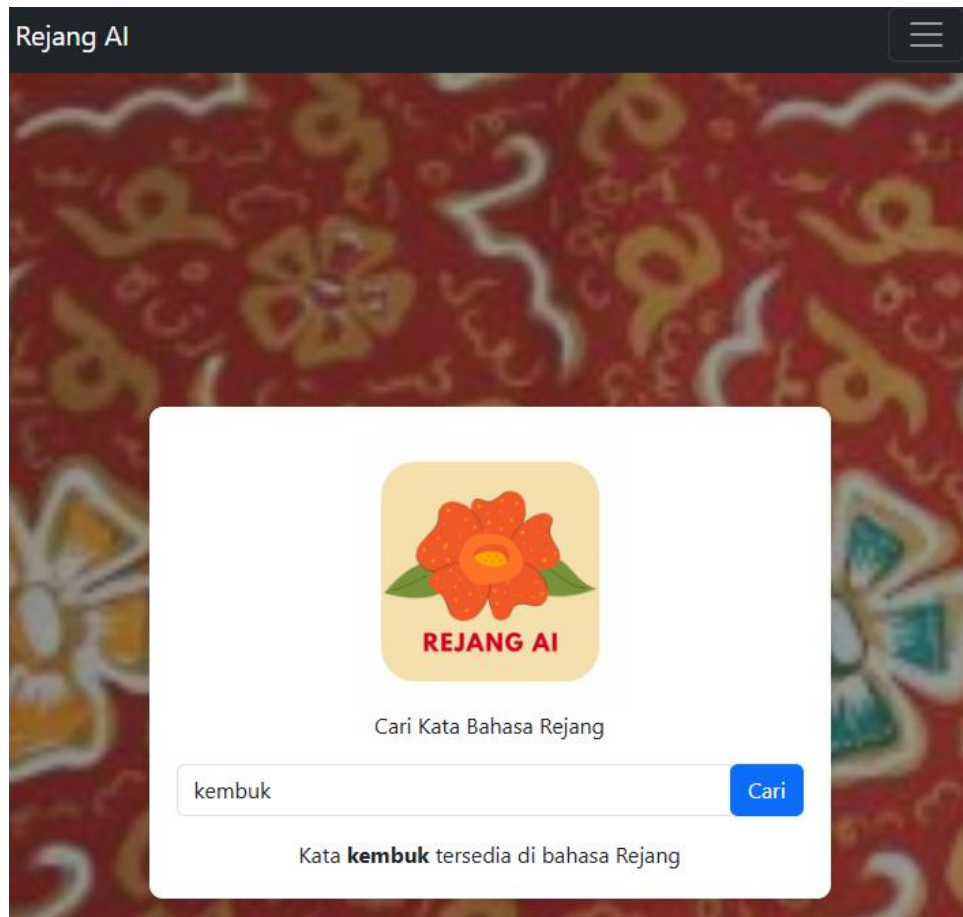


2) History

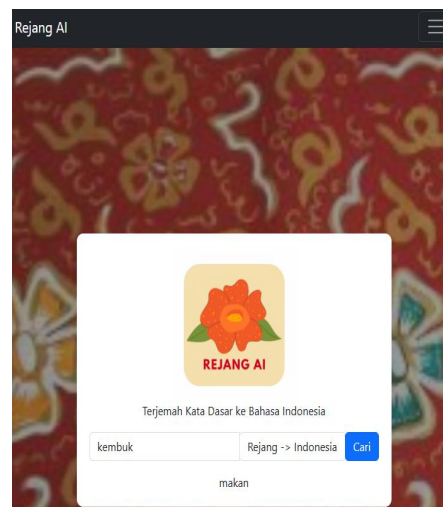
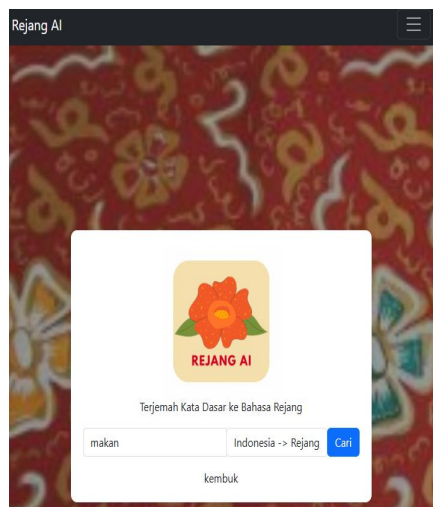


Indonesia dikenal sebagai negara yang memiliki banyak sekali suku bangsa, salah satunya Suku Rejang. Suku Rejang merupakan salah satu suku tertua di Sumatera yang mendiami sebagian wilayah Provinsi Bengkulu saat ini. Sebaran masyarakat Suku Rejang ada di Kabupaten Rejang Lebong, Kabupaten Kepahiang, Bengkulu Tengah, Bengkulu Utara, dan Kabupaten Lebong. Suku Rejang diyakini sebagai penduduk asli wilayah Bengkulu, sekaligus penghuni pertama atau suku tertua.

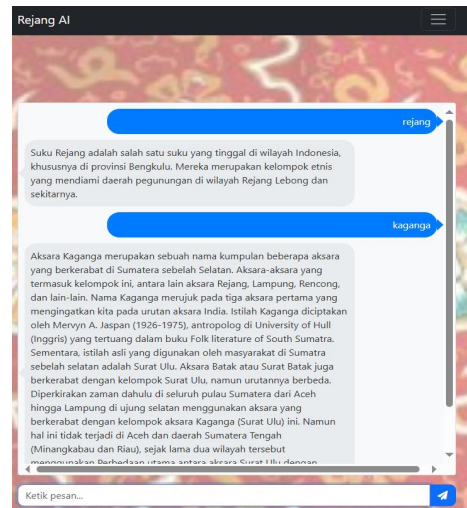
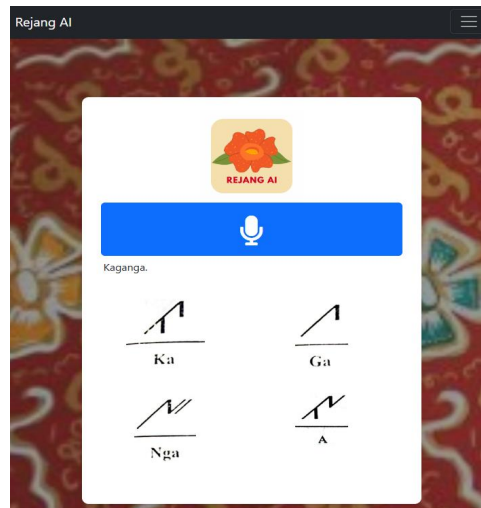
3) Dictionary



4) Translation



5) AI Features



B. Source Code

1) Header

```
<?php include "nav.php ";? >

<div class="container">
<div align="center" class="mt-4" style="font-size:20pt">
Rejang AI is an application to learn about the Rejang language
and culture found in Bengkulu Province.

</div>
</div>

<!-- Bootstrap JS and dependencies (jQuery not needed in
Bootstrap 5) -->
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-
alpha1/dist/js/bootstrap.bundle.min.js"></script>
```

2) History

```
<?php include "nav.php ";? >

<div class="container mt-5">
<?php
$i=mysqli_fetch_array( mysqli_query($conn,"SELECT * from
history where id='1'"));
echo $i['isi'];
?>
</div>

<!-- Bootstrap JS and dependencies (jQuery not needed in
Bootstrap 5) -->
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-
alpha1/dist/js/bootstrap.bundle.min.js"></script>
```

3) Dictionary

```
<?php

// Connection to database
$host = 'localhost';
$dbname = 'beng5632_rejang';
$username = 'beng5632_sastia';
$password = 'sastia@rejang';

$conn = mysqli_connect( $host, $username, $password,
$dbname) ;
```

```
$word = $_GET['word'];
$cmd = mysqli_query( $conn, "SELECT * FROM
katadasar_rejang WHERE kata = '$kata'");

if (mysqli_num_rows($cmd) > 0) {
echo 'The word <b> ' . $ word . '</b> is available in Rejang
language';
} else {
// If the word is not found, search for similar words using
SOUNDEX
$similar_cmd = mysqli_query( $conn, "SELECT kata FROM
katadasar_rejang WHERE SOUNDEX(kata) = SOUNDEX('$kata')
LIMIT 3");

if (mysqli_num_rows($similar_cmd) > 0) {
$similar_words = [];
while ($row = mysqli_fetch_assoc($similar_cmd)) {
$similar_words[ ] = $row['words'];
}
echo 'The word <b> ' . $ word . '</b> is not available in
Rejang. Perhaps you meant: ' . implode( ' ', $similar_words);
} else {
echo 'The word <b> ' . $ word . '</b> is not available in Rejang
and no similar words were found.';
}
}
?>
```

4) Translation

```
<?php include "nav.php ";? >
<style>
body {
background-color: #333;
}
</style>

<div style=" position:relative ; height:90vh; display:flex;
justify-content:center; align-items:center;">

<!-- Pseudo-element for background with opacity -->
<div style="content:";
position:absolute ;
top:0;
left:0;
width:100%;
height:100%;
background-
image:url(https://osccdn.medcom.id/images/content/2021/10/2
6/e22342d2e27e3d8fed17bcb4b88daef6.jpg);
```

```
background-repeat:no-repeat;
background-size:cover ;
opacity:0.5;
z-index:- 1;">
```

```
</div>
```

```
<!-- Form container -->
<div class="container" align="center" style="background-
color:rgba (255, 255, 255, 1);
padding:20px;
border-radius:10px;
box-shadow:0px 4px 10px rgba( 0, 0, 0, 0.1);
max-width:500px;
width:100%;">
<br>
<label>Translate Basic Words into Rejang
Language</span></label>
<div class="input-group mt-3 mb-3">
<input type="search" class="form-control" id="cek"
placeholder="Enter root word">
<div class="input-group-append">
<select id="type" class="form-control" onchange=" ketcx( )">
<option value="torj">Indonesia -> Rejang</option>
<option value="toid">Rejang -> Indonesia</option>
</select>
</div>
<div class="input-group-append">
<a class="btn btn-primary" onclick=" caritlx( )" id="btncek">
Search</a>
</div>
</div>
```

```
<div id="result"></div>
</div>
</div>
```

```
<!-- Bootstrap JS and dependencies (jQuery not needed in
Bootstrap 5) -->
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-
alpha1/dist/js/bootstrap.bundle.min.js"></script>
<!-- jQuery (for AJAX) -->
<script src="https://code.jquery.com/jquery-
3.6.0.min.js"></script>
<script>
$(document ).ready (function(){
    ketcx( ) ;
})
function caritlx( ){
var check = $('#check' ).val ();
```



```

var type= $('#type').val ();
$('#results').html( 'Searching...');
$.ajax ({
    url : 'tlx.php',
method: 'get',
    data : {word:check,type:type},
    success:function (data){
$('#results').html(data);
}
})
}

function ketcx( ){
var type= $('#type').val ();
var ket= $('#ket');
if(type=='torj')
{
ket.html('Rejang');
}
else
{
ket.html('Indonesia')
}
}
</script>
</body>

</html>

```

5) AI Features: Chatbot

```

<?php
// Connection to database
$host = 'localhost';
$dbname = 'beng5632_rejang';
$username = 'beng5632_sastia';
$password = 'sastia@rejang';

// Create connection
$conn = new mysqli( $host, $username, $password, $dbname);

// Check connection
if ($conn->connect_error) {
    die( "Connection failed: " . $conn->connect_error);
}

// Function to search for responses based on keywords
function getResponse( $conn, $keywords) {
// Sort keywords from longest to shortest
    usort( $keywords, function($a, $b) {
return strlen($b) - strlen($a);

```

```

});

// Check each keyword in the database
foreach ($keywords as $keyword) {
    $keyword = $conn->real_escape_string($keyword);
    $query = "SELECT response FROM chatbot WHERE keyword
    LIKE '%$keyword%'";
    $result = $conn->query($query);

    if ($result->num_rows > 0) {
        $row = $result->fetch_assoc();
        return $row['response'];
    }
}

// If no match is found, return the default response
return "Sorry, I don't have any information about that yet.";
}

// Get message from request
if (isset($_POST['message'])) {
    $userMessage = $conn-
    >real_escape_string($_POST['message']);

    // Save user message to database
    $query = "INSERT INTO messages (message, sender) VALUES
    ('$userMessage', 'user')";
    $conn->query($query);

    // Create a list of keywords
    $keywords = explode(' ', strtolower($userMessage)); // Split by
    spaces
    $keywords[ ] = strtolower($userMessage); // Add the entire
    message as keywords

    // Search for responses based on keywords
    $response = getResponse( $conn, $keywords);

    // Save the response from the chatbot to the database
    $query = "INSERT INTO messages (message, sender) VALUES
    ('$response', 'bot')";
    $conn->query($query);

    // Send response back to user
    echo $response;
}

$conn-> close( ) ;
?>

```

6) AI Features: Kaganga

```

<!DOCTYPE html>
<html lang="en">

<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-
scale=1">
<title>Speech Recognition and Database Search</title>

  <!-- Link to Font Awesome for microphone icon -->
<link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/b
otstrap.min.css" rel="stylesheet"
integrity="sha384-
QWTKZyjpPEjISv5WaRU9OFeRpok6YctnYmDr5pNlyT2bRjXh0JMH
jY6hW+ALEwIH" crossorigin="anonymous">
  <!-- Add jQuery from CDN -->
<script src="https://code.jquery.com/jquery-
3.6.0.min.js"></script>
<script
src="https://cdn.jsdelivr.net/npm/@popperjs/core@2.11.8/dist/
umd/popper.min.js"
integrity="sha384-
I7E8VVD/ismYTF4hNIPjVp/Zjvgyol6VFvRkX/vR+Vc4jQkC+hVqc2
pM8ODewa9r" crossorigin="anonymous">
</script>
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/boot
strap.min.js"
integrity="sha384-
0pUGZvbkm6XF6gxjEnlmuGrJXVbNuzT9qBBavbLwCsOGabYfZo0
T0to5eqruptLy" crossorigin="anonymous">
</script>

  <!-- Add styling to spruce up the look a bit -->
<style>
body {
background-color: #333;
}

#micIcon {
font-size: 48px;
cursor: pointer;
color: #333;
}

```

```
#micState {
font-size: 48px;
cursor: pointer;
color: #333;
}

.listening {
color: green;
animation: pulse 1s infinite;
}

@keyframes pulse {
0% {
transform: scale( 1);
}

50% {
transform: scale( 1.1);
}

100% {
transform: scale( 1);
}
}

# result {
resize: none;
}

#searchResults {
margin-top: 20px;
flex: 1;
/* Search results area fills the remaining space */
}

.result -item {
display: flex;
align-items: center;
}

.result -text {
margin-right: 20px;
flex: 1;
/* So that the search result text does not exceed the image
width */
}

.result -image {
width: 100%;
/* To make the image responsive */
}
```

```
height: auto;
}
```

```
@media (max-width: 600px) {
.container {
flex-direction: column;
/* Vertical stack on small screens */
}
```

```
# result {
width: 100%;
/* Input area width 100% on small screens */
}
```

```
.result-item {
flex-direction: column;
/* Vertical stack on small screens */
align-items: flex-start;
}
```

```
.result-image {
max-width: 100%;
/* The image will fill 100% of the container on small screens */
}
}
</style>
```

```
</head>
```

```
<body>
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
<div class="container">
<a class="navbar-brand" href="#">Rejang AI</a>
<button class="navbar-toggler" type="button" data-bs-
toggle="collapse" data-bs-target="#navbarNav" aria-
controls="navbarNav" aria-expanded="false" aria-label="Toggle
navigation">
<span class="navbar-toggler-icon"></span>
</button>
<div class="collapse navbar-collapse" id="navbarNav">
<ul class="navbar-nav mr-auto">
<li class="nav-item">
<a class="nav-link" href= ". / "> History</a>
</li>
<li class="nav-item">
<a class="nav-link" href="dictionary">Dictionary</a>
</li>
<li class="nav-item">
<a class="nav-link" href="translate">Translate</a>
</li>
```

```

<li class="nav-item dropdown">
  <a class="nav-link dropdown-toggle" href="#"
  id="navbarDropdown" role="button" data-bs-
  toggle="dropdown" aria-expanded="false">
  AI Features
  </a>
  <ul class="dropdown-menu" aria-
  labelledby="navbarDropdown">
    <li><a class="dropdown-item"
    href="speak">Kaganga</a></li>
    <li><a class="dropdown-item" href="chat">Chatbot</a></li>
  </ul>
</li>
</ul>
</div>
</div>
</nav>
<div style="position:relative ; height:90vh; display:flex;
justify-content:center; align-items:center;">

  <!-- Pseudo-element for background with opacity -->
  <div style="content:'';
    position:absolute ;
    top:0;
    left:0;
    width:100%;
    height:100%;
    background-
    image:url(https://osccdn.medcom.id/images/content/2021/10/2
    6/e22342d2e27e3d8fed17bcb4b88daef6.jpg);
    background-repeat:no-repeat;
    background-size:cover ;
    opacity:0.5;
    z-index:- 1;">

</div>

  <!-- Form container -->
  <div class="container" align="center" style="background-
  color:rgba (255, 255, 255, 1);
  padding:20px;
  border-radius:10px;
  box-shadow:0px 4px 10px rgba( 0, 0, 0, 0.1);
  max-width:500px;
  width:100%;">
    <br>
    <div align="center" class="mx-2">
      <div class="btn btn-primary w-100" id="micIcon">
        <i class="fas fa-microphone text-white"></i>
      </div>
    </div>
  </div>

```

```
<div class="btn btn-warning w-100 btn-disabled" id="micState"
style="display: none;">
<i class="fas fa-headphones text-dark fa-fade"> </i>
</div>
</div>
```

```
<!-- Area to display the recording results -->
<textarea id="result" class="form-control" placeholder="The
recording results will appear here..." rows="1"
style="border:0px"></textarea>
```

```
<div class="mt-4 row pb-4" id="searchResults " align
="center"></div>
```

```
</div>
</div>
```

```
<script>
$(document ).ready (function() {
// Check if the browser supports the Web Speech API
if ('webkitSpeechRecognition' in window) {
const recognition = new webkitSpeechRecognition( );
recognition.continuous = false; // Stop after a single input
recognition.interimResults = false; // Don't show interim
results
recognition.lang = 'id-ID'; // Set language to Indonesian
```

```
const micIcon = $('#micIcon');
const micState = $('#micState');
const resultTextarea = $('#result');
const searchResults = $('#searchResults');
```

```
// When click microphone icon, start listening
micIcon.on('click', function( ) {
recognition.start ( );
micIcon.addClass('listening'); // Add animation
micIcon.hide(); // Hide the microphone icon
micState.show(); // Show the headphone icon
});
```

```
// When voice recognition starts
recognition.onstart = function() {
console.log( "Listening...");
};
```

```
// When speech recognition successfully captures sound
recognition.onresult = function(event) {
const transcript = event.results [0][0].transcript;
resultTextarea.val(transcript); // Display the results
```

```
const pattern = extractPattern(transcript); // Extract pattern
from recording
searchInDatabase(pattern); // Call the search function with the
pattern
};
```

```
// When voice recognition ends
recognition.onend = function() {
micIcon.removeClass('listening'); // Remove animation
micIcon.show(); // Show the microphone icon again
micState.hide(); // Hide the headphone icon
console.log( "Voice recognition stopped.");
};
```

```
// Function to extract patterns from the recording results
function extractPattern(transcript) {
// Take the first pattern from the recorded result or at least one
character
const pattern = transcript.trim (); // Remove extra spaces
return pattern.length > 0 ? pattern : null;
}
```

```
// Function to search in database using AJAX jQuery
function searchInDatabase(query) {
$.ajax ({
url: 'search.php',
type: 'GET',
data: {
query: query
},
success: function(response) {
// If the data is a JSON string, parsing is required
const data = typeof response === 'string ' ?
JSON.parse(response) : response;
displayResults(data); // Display results if JSON is valid
},
error: function( xhr, status, error) {
console.error ('Error:', error); // Debugging if an
error occurs
}
});
}
```

```
// Function to display search results
function displayResults(data) {
searchResults.empty(); // Empty previous results

if ( data.length > 0) {
// Create a row to display the results in a two-column grid
```



```

const row = $('<div>').addClass ('row');

    $.each (data, function(index, item) {
// Create a col-6 div element to create a two-column layout
const col = $('<div>').addClass ('col-6 mb-3');

// Create an image element if an image URL is available
if ( item.image ) {
const img = $('<img>').attr ('src', 'kaganga/' + item.image)
                    .attr ('alt', `Result image for ${item.word}`)
                    .addClass ('img-fluid p-2'); // Add img-fluid
class for responsiveness

// Create a percentage text element
const percentageText = $('<p>').text( `Match:
${item.percentage}%`)
                    .addClass ('text-muted'); // Text is grayed
out

// Add image and percentage to the column
col.append (img);
// col.append (percentageText);
}

    row.append (col); // Add a column to the row
});

searchResults.append(row); // Add the result row to the DOM
} else {
searchResults.text('No results found.');
```

```

}
}

} else {
    alert( "Your browser does not support the Web Speech
API");
}
});
</script>
</body>
</html>

```

C. Program Usage

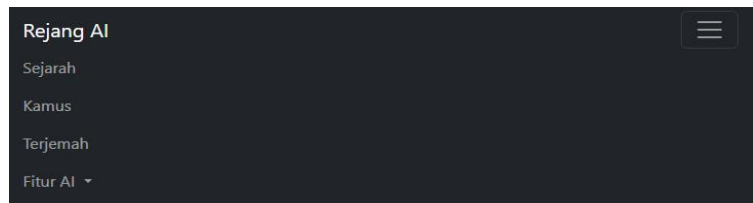
The application "A COMBINATION OF REJANG AND KNUTH MORRIS STEMMING ALGORITHM FOR AI-BASED INDONESIAN REJANG LANGUAGE TRANSLATOR" or called REJANG AI, is an application designed to help people get to know and learn the Rejang language. Rejang language is a regional language found in Bengkulu Province, Rejang language is included in the Malay language group in the Sumatran archipelago. This application is equipped with menus: History, Digital Dictionary, Translation and AI Features namely Chatbot and Voice Recognition (Kaganga). To be able to use or run the REJANG AI application, it can be accessed via the link <https://rejang-ai.my.id/>. The REJANG AI application is designed in a simple and attractive interface so that users can interact easily, as for how to use or run the REJANG AI application, as follows:

1) Home

This is the main page of the REJANG AI application.

How to use:

- Open your browser and type <https://rejang-ai.my.id/>



Rejang AI merupakan aplikasi untuk mengenal tentang Bahasa dan Budaya Rejang yang terdapat di Provinsi Bengkulu



2) History

This is a page that displays a historical interface regarding the Rejang language.

How to use:

- Open your browser and type <https://rejang-ai.my.id/>
- Then click the "History" submenu
- The results will be displayed as below:



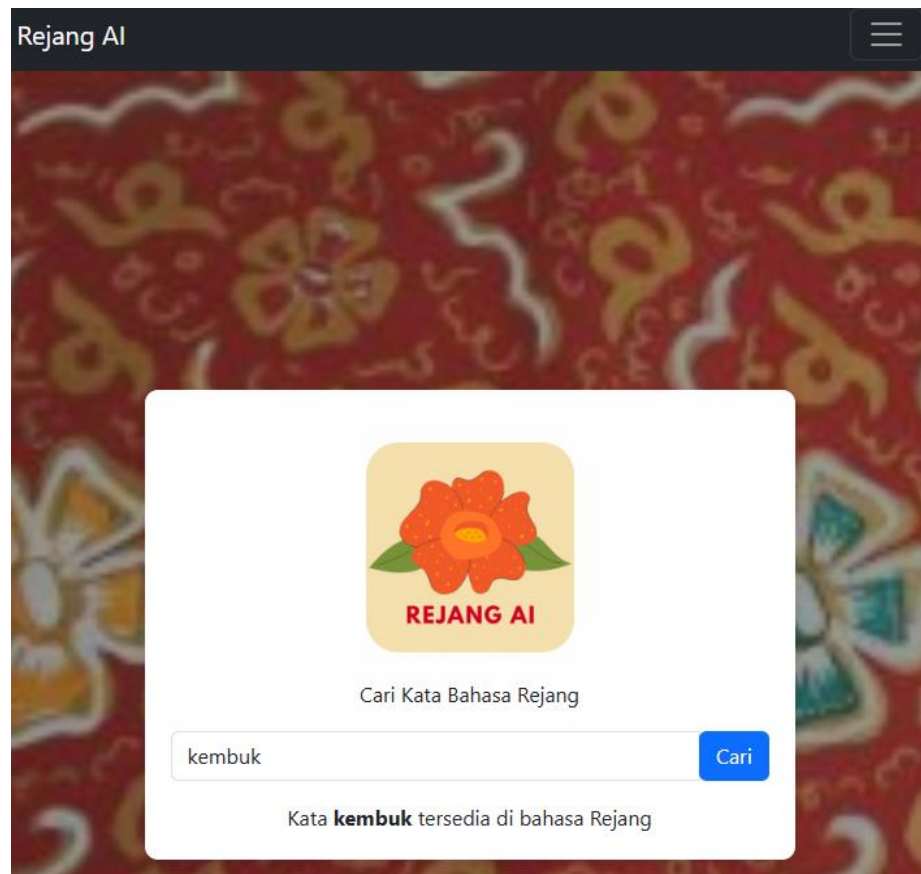
Indonesia dikenal sebagai negara yang memiliki banyak sekali suku bangsa, salah satunya Suku Rejang. Suku Rejang merupakan salah satu suku tertua di Sumatera yang mendiami sebagian wilayah Provinsi Bengkulu saat ini. Sebaran masyarakat Suku Rejang ada di Kabupaten Rejang Lebong, Kabupaten Kepahiang, Bengkulu Tengah, Bengkulu Utara, dan Kabupaten Lebong. Suku Rejang diyakini sebagai penduduk asli wilayah Bengkulu, sekaligus penghuni pertama atau suku tertua.

3) Dictionary

This is a page that displays the Rejang digital dictionary interface.

How to use:

- Open your browser and type <https://rejang-ai.my.id/>
- Then click the "Dictionary" submenu
- Type the word Rejang, for example "kembuk"
- Then click "Search"
- The results will be displayed as below:

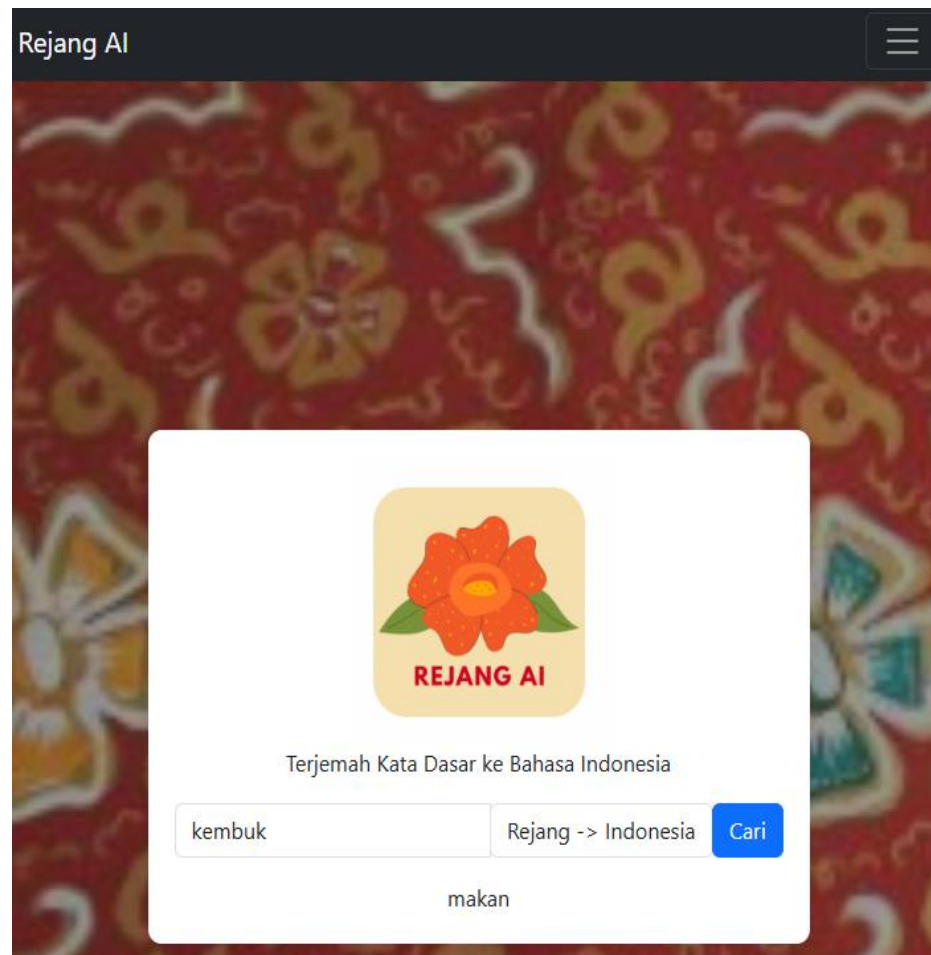


4) Translation

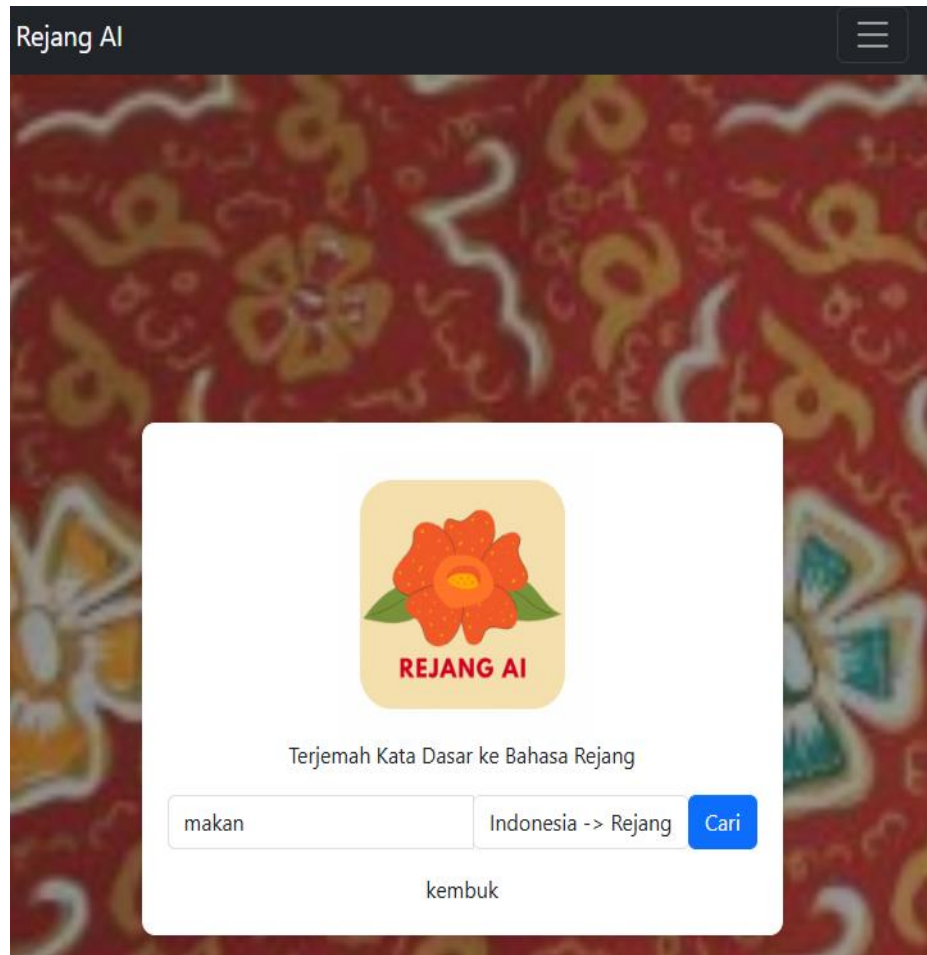
This is a page that displays the translation interface from Rejang to Indonesian or vice versa.

How to use:

- Open your browser and type <https://rejang-ai.my.id/>
- Select the "Translate" submenu
- Click on the "Rejang →Indonesia" section for translation settings, for example Rejang to Indonesia
- Type the word Rejang, for example "kembuk"
- Then click "Search"
- The results will be displayed as below:



- For Indonesian to Rejang translation, simply change the settings in "Indonesia → Rejang", and type an Indonesian word such as "makan", the result will appear as below:

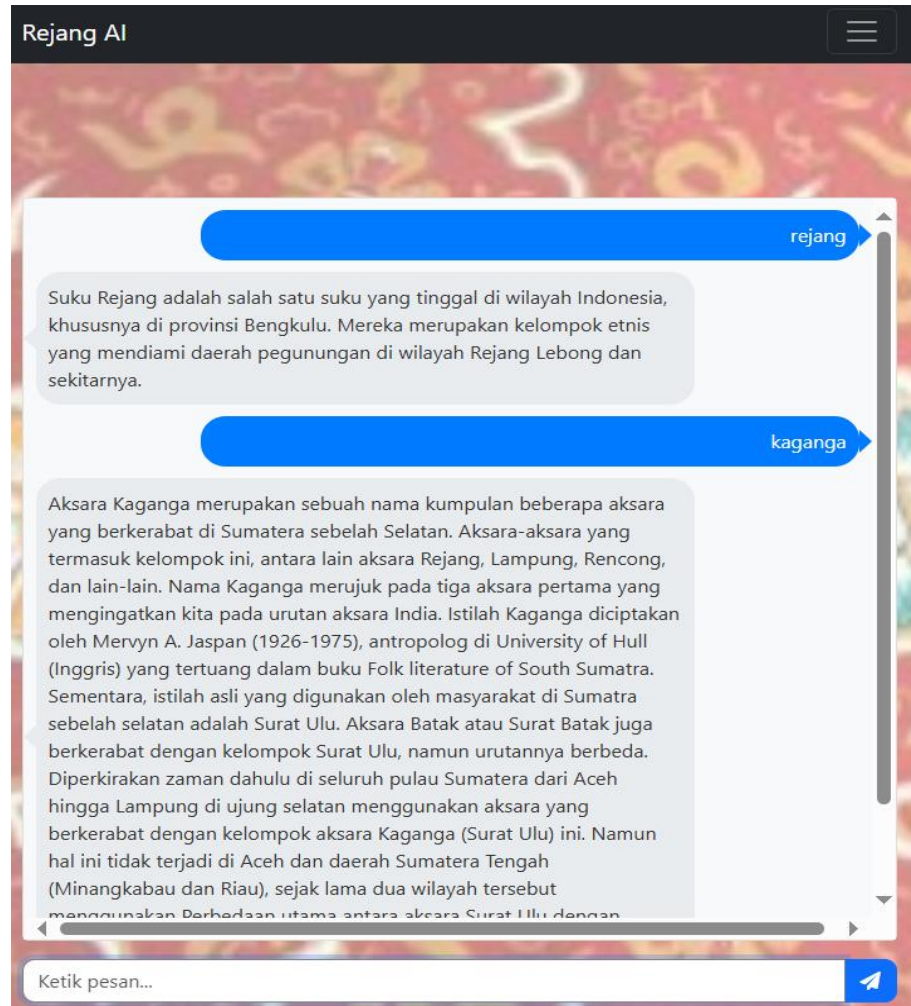


5) AI Features: Chatbot

This is a page that displays the Chatbot interface which functions for interaction between users and applications.

How to use:

- Open your browser and type <https://rejang-ai.my.id/>
- Select the "AI Features" submenu
- Select "Chatbot"
- Type keywords related to Rejang, for example "Rejang Tribe"
- The results will be displayed as below:

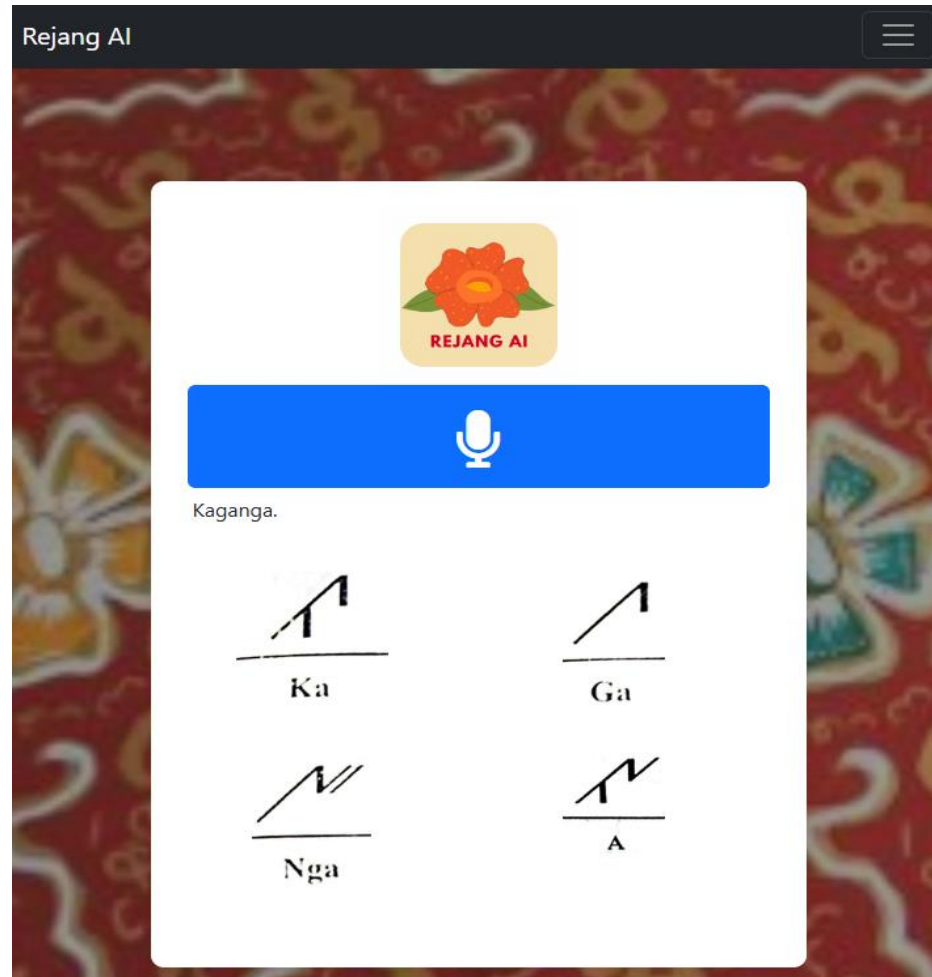


6) AI Features: Kaganga

This is a page that displays the Voice Recognition interface which functions to display Kaganga letters with sound patterns.

How to use:

- Open your browser and type <https://rejang-ai.my.id/>
- Select the "AI Features" submenu
- Choose Kaganga
- Say one of the Kaganga letters, for example "kaganga"
- The results will be displayed as below:



D. Data Input and Editing

1) History

Adding history can be done via the link <https://rejang-ai.my.id/sejarahdata>. This link serves to add information about Rejang's history. Here's how it looks:

Edit Sejarah

Isi Sejarah



Simpan Perubahan

2) Data Dictionary

You can input and edit the Data Dictionary through the link <https://rejang-ai.my.id/kamusdata>. This link allows you to enter, edit, and delete Rejang and Indonesian vocabulary. Here's how it looks:

Rejang AI Data Sejarah Data Kamus Data Chatbot			
Tambah Kata			
No	Rejang	Indonesia	Aksi
1	a lau	enyah	<button>Edit</button> <button>Hapus</button>
2	a'ang	arang	<button>Edit</button> <button>Hapus</button>
3	aba	aba	<button>Edit</button> <button>Hapus</button>
4	abad	abad	<button>Edit</button> <button>Hapus</button>
5	abadi	abadi	<button>Edit</button> <button>Hapus</button>
6	abaei	lalai	<button>Edit</button> <button>Hapus</button>
7	abai	mengabaikan	<button>Edit</button> <button>Hapus</button>
8	abak	papa	<button>Edit</button> <button>Hapus</button>
9	abak	bapak	<button>Edit</button> <button>Hapus</button>
10	abas	hilang	<button>Edit</button> <button>Hapus</button>

Tambah Kata Terjemah



Kata Rejang

Kata Indonesia

Simpan

Edit Kata Terjemah



Kata Rejang

abak

Kata Indonesia

bapak

Simpan Perubahan

rejang-ai.my.id says

Apakah Anda yakin ingin menghapus data ini?

OK

Cancel

3) Chat Data

To add a chatbot, go to <https://rejang-ai.my.id/chatdata>. This link allows you to add keywords to the chatbot. Here's what it looks like:

[Data Sejarah](#) [Data Kamus](#) [Data Chatbot](#)

[Tambah Keyword](#)

Keywords : Sejarah
Rejang merupakan suku yang penduduknya mayoritas di provinsi Bengkulu

Hapus

Keywords : rejang
Suku Rejang adalah salah satu suku yang tinggal di wilayah Indonesia, khususnya di provinsi Bengkulu. Mereka merupakan kelompok etnis yang mendiami daerah pegunungan di wilayah Rejang Lebong dan sekitarnya.

Hapus

Keywords : bahasa
Suku Rejang memiliki bahasa sendiri yang termasuk dalam kelompok bahasa Melayu-Polinesia. Bahasa Rejang masih digunakan dalam kehidupan sehari-hari oleh masyarakat mereka.

Hapus

Tambah Keyword



Keyword

Bahasa

Response

Suku Rejang memiliki bahasa sendiri yang termasuk dalam kelompok bahasa Melayu-Polinesia. Bahasa Rejang masih digunakan dalam kehidupan sehari-hari oleh masyarakat mereka.

Simpan

